

# **From Cross-Sections to Dynamics**

## Session 5

---

Ethan Ligon

September 15, 2026

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

## The problem

---

Ghana's GLSS gives seven rounds, but almost never the same household twice. We want

$$\Delta \log c_{it} = \alpha_i + \delta_t + \beta' \Delta z_{it} + \epsilon_{it},$$

and  $\alpha_i$  is not identified because  $i$  appears once.

Deaton's answer: give up on households and follow *cohorts*. A cohort — say, everyone born 1965–69 — is present in every round, and its membership is fixed by construction.

Define cohort  $c$  by birth year of the head. For each  $(c, t)$  compute the weighted mean  $\bar{y}_{ct}$ . Then

$$\bar{y}_{ct} = \alpha_c + \delta_t + \beta' \bar{z}_{ct} + \bar{\epsilon}_{ct}$$

is a genuine panel, in  $C \times T$  cells, and  $\alpha_c$  is estimable.

The catch:  $\bar{y}_{ct}$  is a *sample* mean, so it carries measurement error of order  $1/\sqrt{n_{ct}}$ . With  $\bar{z}$  on the right-hand side that is classical errors-in-variables, and  $\hat{\beta}$  attenuates.

## Identification: age, cohort, and time

---

You cannot have all three. Since

$$\text{age} = \text{year} - \text{birth year},$$

age, period, and cohort effects are exactly collinear.

Something must be restricted:

- drop period effects (a "pure lifecycle" reading), or
- restrict period effects to sum to zero and be orthogonal to trend (Deaton & Paxson's normalization), or
- bring in outside information on one of the three

This is not a technical nuisance to be worked around. It is a statement that the data cannot answer the question without an assumption, and the assumption should be argued for.

## A genuine panel: Uganda

---

The Uganda National Panel Survey follows households from 2005–06 to 2018–19. With the same  $i$  in multiple  $t$ :

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}$$

is estimable by within transformation.

- $\alpha_i$  absorbs everything time-invariant — soil quality, ability, the district's road
- what remains identified is the effect of variation *within* the household over time
- and  $\beta$  is now a within estimate, which is a different parameter from the cross-sectional one, not a better-identified version of it

# Inference

---

Cluster at the level of treatment assignment, not at the level of sampling.

- sampling clusters (EAs): cluster the SEs, per session 1
- if the variation you exploit is at district level, cluster on district
- few clusters ( $G < 40$ )  $\Rightarrow$  wild cluster bootstrap
- two-way (household and time) when  $T$  is not small

Abadie et al. (2023): clustering is about *design* or *sampling*, and if neither applies, clustering is not conservative — it is wrong.

$\rightarrow$  notebook



## Exercises

---

1. Rebuild the Ghana pseudo-panel with ten-year cohorts. How much do the lifecycle profiles change? Which conclusion is robust?
2. Implement the Verbeek–Nijman errors-in-variables correction using the within-cell variances, and report the corrected  $\beta$ .
3. In Uganda, quantify attrition: what fraction of the 2009–10 households appear in 2011–12? Are the leavers different in 2009–10 consumption from the stayers? What does that do to your fixed-effects estimate?